

HECTOR GENARO PACHECO

Software Engineer | Full-Stack • Cloud • AI • IoT • Embedded Systems

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PROFESSIONAL SUMMARY

Software Engineer with a physical-world engineering background spanning full-stack development, AWS cloud infrastructure, IoT edge computing, AI/data platforms, and industrial instrumentation & controls. Builds auditable software systems ranging from containerized AWS applications and GraphQL business platforms to Raspberry Pi telemetry and hardware-integrated software.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, SQL, C++

Frontend: React, Next.js, HTML5, CSS3

Backend & APIs: Node.js, Express, GraphQL, REST APIs, FastAPI, WebSockets, Zod

Embedded & Industrial: Raspberry Pi 5, GPIO, GPS/NMEA, Sensors, 4–20 mA Loops, PLC I/O

Cloud & Infra: AWS ECS Fargate, ECR, RDS, Secrets Mgr, KMS, VPC, Docker, Vercel, Linux, Git

Data: PostgreSQL, MongoDB, Prisma, Supabase, Apollo

AI & Data Platforms: Palantir Foundry, Palantir AIP, AI Workflow Training, Data Governance

FEATURED DEPLOYED & ENGINEERING PROJECTS

CalTrack — Industrial Calibration SaaS & Cloud Platform [LIVE APP ↗](#) [GITHUB ↗](#) React • TypeScript • Node • Express • PostgreSQL • AWS ECS Fargate

- Architected and deployed a full-stack industrial calibration platform for instrument tags, calibration records, authentication/RBAC, and persistent audit histories.
- Containerized Node/Express API with Docker and deployed it using AWS ECS Fargate, ECR, RDS PostgreSQL, Secrets Manager, KMS, and Vercel.

Procurement Tracking System — Enterprise Procurement Workflow [LIVE APP ↗](#) [GITHUB ↗](#) React • TypeScript • GraphQL • Apollo • Express • Prisma • PostgreSQL

- Built a deployed procurement operations platform supporting purchase requests, approval workflows, vendor management, purchase orders, goods receiving, and multi-role RBAC.
- Implemented GraphQL APIs with Apollo Server/Client, Prisma/PostgreSQL persistence, JWT authentication, and backend resolver permission enforcement.

FieldTrack AI — Edge AI & Sensor Telemetry Platform [GITHUB ↗](#) Raspberry Pi 5 • Python • FastAPI • Node • React • WebSockets

- Built an edge-computing platform combining Python hardware services, GPS/NMEA parsing, PIR motion sensing, camera/computer-vision processing, and real-time React telemetry over WebSockets.

Thriveward Funding Intelligence — AI-Assisted Funding Intelligence Platform [LIVE APP ↗](#) [GITHUB ↗](#) React • TypeScript • Express • Prisma • PostgreSQL • Docker

- Built provenance-aware funding-opportunity workflows with structured ingestion, source traceability, persistent data modeling, Argon2id authentication, RBAC, auditability, and human review controls.

PRODUCTION CLIENT WORK & ADDITIONAL PROJECTS

OC Water Features — Production Full-Stack Web Platform [LIVE SITE ↗](#) Next.js • TypeScript • Supabase PostgreSQL • Zod • Vercel

- Built and deployed a production web platform with SEO, lead intake workflows, customer-facing forms, administrative tooling, and persistent database-backed workflows.

Additional Engineering Repositories: **Pi Arcade OS** — Embedded Hardware & Software Gaming Platform (Raspberry Pi / Python / Pygame / GPIO) [PROJECT SITE ↗](#) [GITHUB ↗](#) • **E-Commerce Store** — Full-Stack GraphQL Store (React / GraphQL / MongoDB / AWS S3) [GITHUB ↗](#)

TECHNICAL & INDUSTRIAL EXPERIENCE

Instrumentation & Controls Technician — Process Control & Industrial Systems [GITHUB ↗](#) Field Engineering Experience

- Calibrated, commissioned, and troubleshot 4–20 mA process transmitters (pressure, temperature, flow, level), PLC I/O wiring harnesses, terminal blocks, relays, and field instruments across 480VAC, 120VAC, and 24VDC power systems.
- Verified process control loop operation, signal integrity, and safety shutdowns while maintaining auditable calibration logs.

EDUCATION & PROFESSIONAL TRAINING

Western Governors University (WGU) — B.S. Software Engineering (In Progress, 2026–Present)

Palantir Technologies — Foundry & AIP Training (May 2026)

Mirion Technologies — SU-890 iCAM Operations & Maintenance (March 2026, 16.00 Continued Education Credits)